

DEVOPS

* DevOps Notes *

Let's
Build
Together 😊

1. DEFINITION

DevOps is a combination of "Development" and "Operations".

It is a culture, practice and set of tools that improves an organization's ability to deliver applications and services at high velocity.



Goal → Deliver Value
Faster, Better, Cheaper

2. INTRODUCTION

- Traditional approach had gap between Dev and Ops.
- DevOps breaks these silos.
- Focus on automation, collaboration and continuous improvement.
- Helps in rapid delivery of high-quality software.

3. WHY DEVOPS ?

- ✓ Faster Delivery 🚀
- ✓ Improved Collaboration 👤
- ✓ Better Quality 🛡️
- ✓ Continuous Improvement 📈
- ✓ Efficient Use of Resources ⚙️
- ✓ Customer Satisfaction 😊

REAL-WORLD UNDERSTANDING 💡

EX: Think of a Food Delivery App

- Dev builds new features
- Ops makes sure app is always available to users
- DevOps makes sure new features are released quickly without breaking the app.

= Happy Customer 😊

KEY POINTS

- DevOps is not a tool, it's a Mindset.
- People + Process + Tools = DevOps Success
- Automation is the ❤️ of DevOps.
- It's an ongoing journey, not a destination.

Quick Notes

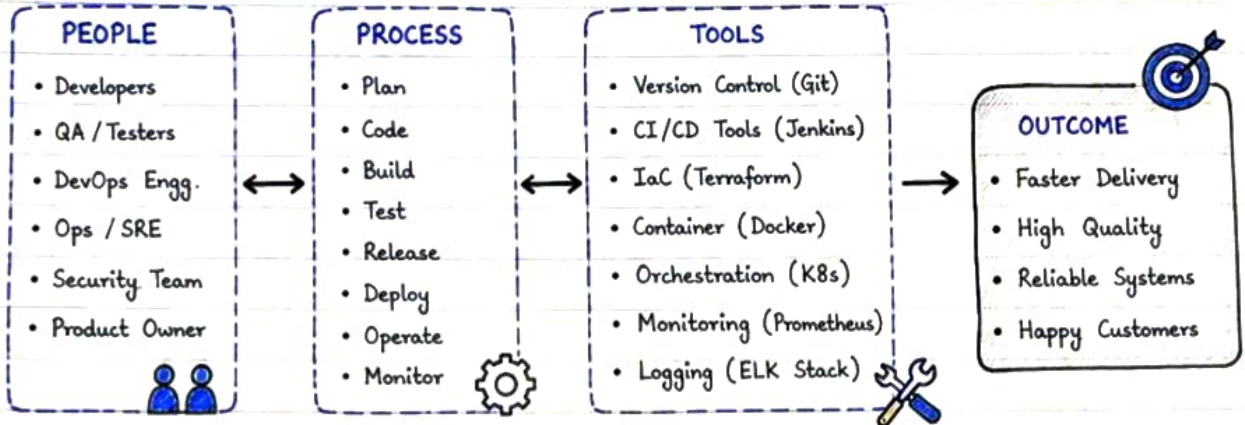
DevOps → Faster Delivery
Better Collaboration
Continuous Improvement

“ Continuous Delivery is the foundation, DevOps is the Culture. ”

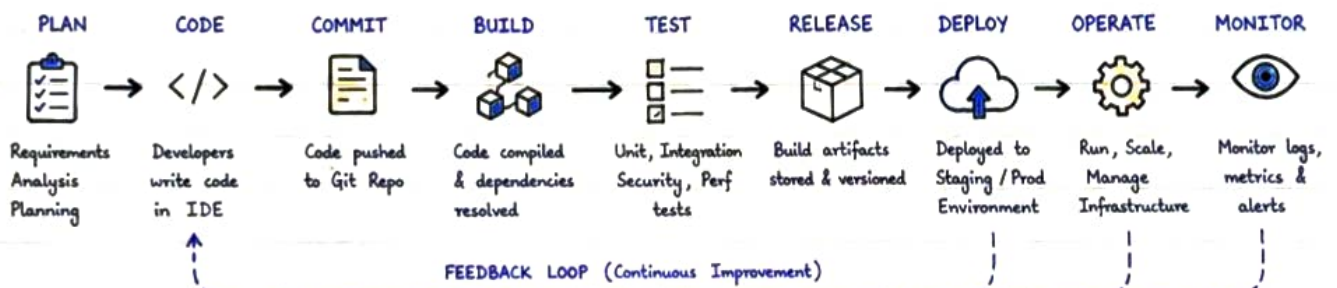
2. DEVOPS ARCHITECTURE

DevOps architecture connects People, Process and Tools to deliver value continuously.

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3. DEVOPS WORKING (CONTINUOUS FLOW)



4. KEY COMPONENTS

- ★ **CI (Continuous Integration)** - Code integrated frequently into shared repo.
- ★ **CD (Continuous Delivery)** - Automated delivery of code to staging environment.
- ★ **IaC (Infrastructure as Code)** - Manage Infra using code (Terraform, Ansible).
- ★ **Monitoring** - Real-time visibility of system health & performance.
- ★ **Logging** - Centralized logs for troubleshooting.
- ★ **Security** - Integrated at every stage (DevSecOps).

ARCHITECTURE LAYERS

	User Layer	Users access app
	Application Layer	Your application
	Automation Layer	CI/CD, IaC, Scripts
	Infrastructure Layer	Servers, Cloud, Network
	Monitoring Layer	Logs, Metrics, Alerts

REAL-WORLD EXAMPLE

Think of Amazon → Code commit → Build → Test → Deploy → Monitor 24/7 across thousands of servers globally. This is DevOps in action!

Quick Notes

DevOps is not just about tools, it's about culture, automation and continuous improvement.



DEVOPS LIFECYCLE

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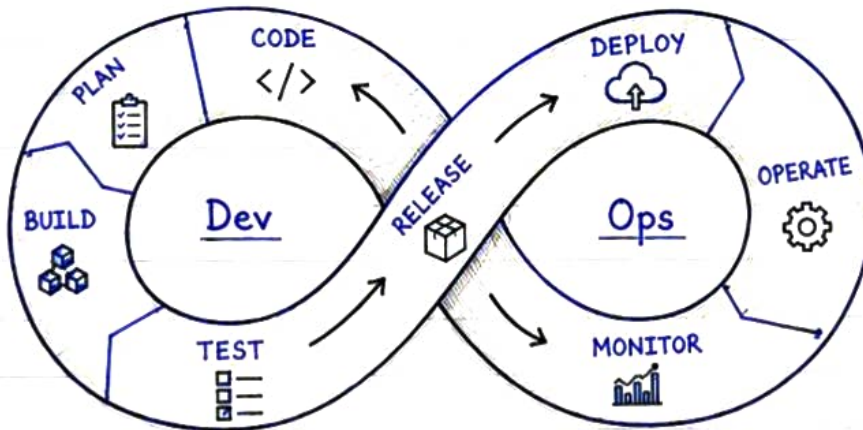
DEFINITION :

DevOps Lifecycle is a continuous process that helps deliver high-quality software faster, more reliably and with continuous improvement.

KEY POINTS

- End-to-End Process
- Continuous Flow
- Collaboration
- Automation
- Feedback Driven

THE INFINITY LOOP



REAL-WORLD UNDERSTANDING

Think of it like a **FOOD ORDER** from planning the menu to delivering it hot to the customer and keeps improving based on feedback.

LIFECYCLE STAGES (1-LINE EXPLANATION)

	1. PLAN	Define <u>requirements</u> , scope, timeline and resources.
	2. CODE	Developers <u>write</u> and commit code into version control system.
	3. BUILD	Code is compiled and dependencies are resolved to create <u>build artifact</u> .
	4. TEST	Automated tests are executed to ensure <u>code quality</u> and functionality.
	5. RELEASE	Approved build is packaged and <u>versioned</u> for deployment.
	6. DEPLOY	Application is deployed to <u>target environment</u> (Staging/Prod).
	7. OPERATE	Application is run, managed and users are served. Ensure <u>reliability</u> and <u>performance</u> .
	8. MONITOR	Monitor system health, lags, metrics and user behavior. Collect <u>feedback</u> for continuous improvement.

QUICK NOTES

DevOps Lifecycle never stops. It's a continuous loop of build, measure and improve.

REMEMBER

- ✓ Automation + Collaboration = Speed ⚡
- ✓ Feedback + Monitoring = Quality ★
- ✓ Dev + Ops Culture = Success 🏆

MOTTO:

Build → Test → Release → Deploy →
Operate → Monitor → Repeat ∞

DEVOPS

Benefits &
Real-World
Example

BENEFITS OF DEVOPS



1. **FASTER DELIVERY** → Code changes reach users quickly through automated pipeline.



2. **BETTER COLLABORATION** → Dev, Ops, QA, Security work together. No more silos.



3. **AUTOMATION** → Reduces manual work, human errors and increases consistency.



4. **CONTINUOUS FEEDBACK** → Get real-time feedback from users, logs, metrics to improve continuously.



5. **SCALABILITY** → Applications can scale up/down based on demand automatically.



6. **RELIABILITY** → Automated tests, monitoring and rollback ensure stable systems.

WHY IMPORTANT ?

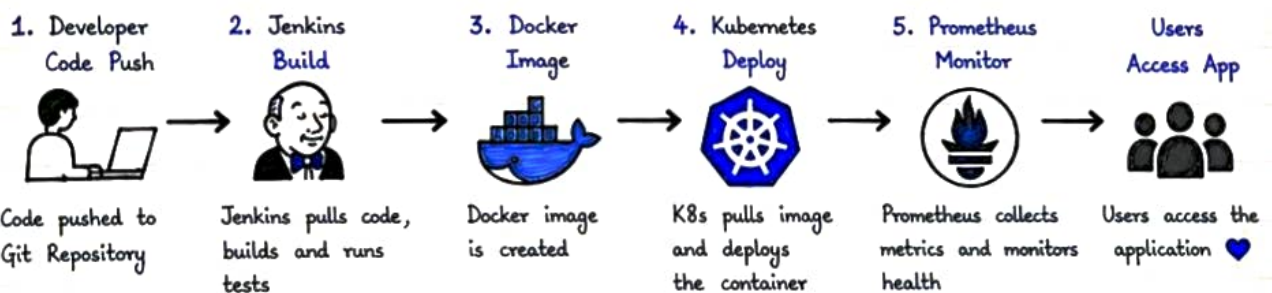
In today's world,
Speed = Competitive
Advantage



QUICK NOTES

- ✓ Happy Team 😊
- ✓ Happy Customer 😊
- ✓ Lower Cost ↓
- ✓ Better Quality ★

REAL-WORLD EXAMPLE (END-TO-END FLOW)



FLOW SUMMARY

Developer code push karta hai → Jenkins build karta hai → Docker image banti hai → Kubernetes deploy karta hai → Prometheus monitor karta hai.

RESULT

- ✓ Fast Delivery
- ✓ High Quality
- ✓ Happy Users



KEY TAKEAWAY

DevOps is not just about tools, it's about mindset, process and people working together to deliver value continuously. ★

“ Automate Everything
Measure Everything
Improve Continuously ”

DEVOPS TOOLS

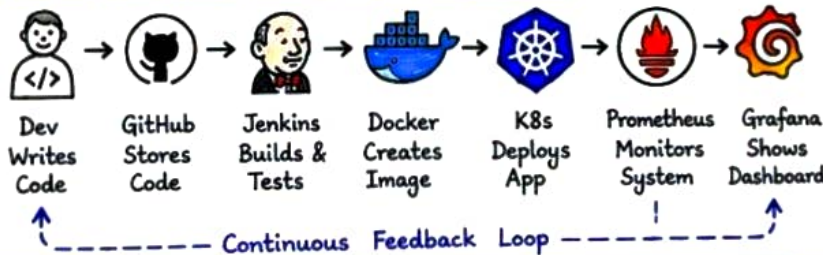
Right Tool
at Right Time
= DevOps Success

DevOps uses different tools to Automate, Build, Deploy, Monitor and Improve continuously.



TOOL	PURPOSE (1-LINE)	CATEGORY
① Git	→ Distributed <u>Version Control</u> System to track changes in code.	Version Control
② GitHub / GitLab	→ Host code online, <u>Collaboration</u> PR, Issues, CI/CD integration.	Code Hosting & Collaboration
③ Jenkins	→ Automate Build, Test and Deploy using <u>Pipelines</u> .	CI/CD Tool
④ Docker	→ Create lightweight <u>Containers</u> to package and run applications.	Containerization
⑤ Kubernetes (K8s)	→ <u>Orchestrate</u> Containers, Auto-scaling, Self-healing, Load balancing.	Container Orchestration
⑥ Terraform	→ <u>Infrastructure as Code (IaC)</u> - Provision Cloud Resources automatically.	IaC / Cloud Provisioning
⑦ Ansible	→ Configuration Management & <u>Automation</u> - Manage servers easily.	Automation & Configuration
⑧ Prometheus	→ Monitor Systems, Collect <u>Metrics</u> , Generate Alerts.	Monitoring & Alerting
⑨ Grafana	→ Visualize Metrics from Prometheus in Beautiful <u>Dashboards</u> .	Visualization & Dashboard

HOW THESE TOOLS WORK TOGETHER ?



QUICK NOTES

- ✓ No Single Tool is Best
- ✓ Combination of Tools Makes DevOps Powerful.
- ✓ Choose Tools based on Project Needs.
- ✓ Learn Basics First, Then Go Deep.

★ KEY TAKEAWAYS

- ✓ Tools are enablers, not the goal.
- ✓ Automation + Right Tools = Faster Delivery.
- ✓ Monitor Everything, Improve Continuously.
- ✓ Keep it Simple, Scalable and Reliable.



REMEMBER

Tools change with time,
But Concepts remain same.

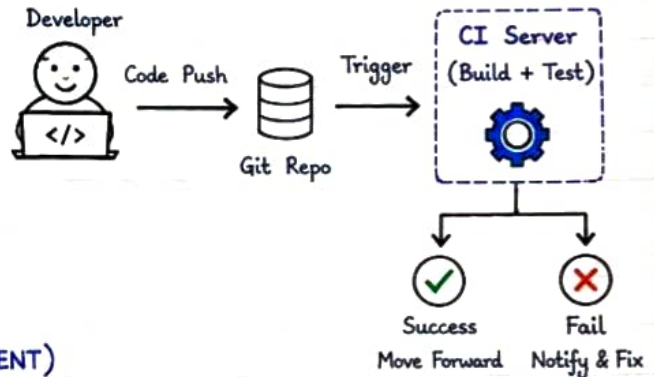


CI / CD in DEVOPS

Automate
Everything
Deliver
Continuously ☺

1. CI (CONTINUOUS INTEGRATION)

- Developers integrate code into shared repository frequently.
- Every integration triggers an automated build and test.
- Goal is to detect issues early and keep codebase stable.



2. CD (CONTINUOUS DELIVERY / DEPLOYMENT)

- After successful testing, code is automatically prepared for release.
- Continuous Delivery → Manual approval before deploy.
- Continuous Deployment → Auto deploy to production.
- Goal is to deliver value to users faster and reliably.

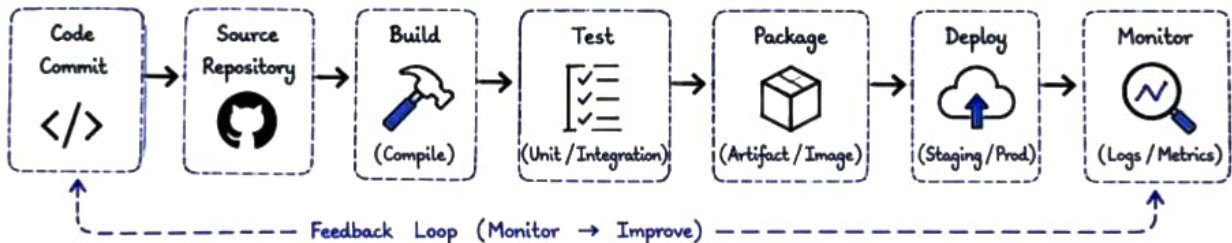
CONTINUOUS DELIVERY (CD)

Code is always in deployable state.
Human approval before production. 🧑

CONTINUOUS DEPLOYMENT (CD)

Code is automatically deployed to production after all tests pass. 🤖

3. SIMPLE CI/CD PIPELINE DIAGRAM



4. SHORT WORKFLOW

- Developer writes code and commits to Git.
- Code pushed to GitHub / GitLab.
- CI server (e.g., Jenkins) pulls code.
- Build is triggered and code is compiled.
- Automated tests run (unit + integration).
- If tests pass → Artifact / Docker image is created.
- CD process deploys to staging / production.
- Monitoring tools check health and performance.
- Feedback collected and cycle repeats.

Automate
→ Build
→ Test
→ Deploy
→ Monitor
→ Repeat
∞

QUICK NOTES

- ✓ CI = Integrate Early, Test Early
- ✓ CD = Deliver Fast, Deploy Often
- ✓ Automation reduces human errors.
- ✓ Small changes, frequent releases = Happy Users ☺
- ✓ Always monitor and improve. ★

★ INTERVIEW TIP

CI focuses on "Integration + Testing"
CD focuses on "Delivery + Deployment"



Together they form the backbone of DevOps.

DEVOPS INTERVIEW QUESTIONS

Prepare
Well
Crack Any
Interview! 📌

1. Q. What is DevOps? **Ans:** DevOps is a culture and set of practices that connects Dev (Development) and Ops (Operations) to deliver software faster and reliably.
2. Q. Why DevOps? **Ans:** To deliver software faster, improve quality, increase collaboration and automate processes.
3. Q. Difference between Agile & DevOps? **Ans:** Agile is about iterative development. DevOps is about automating delivery and operations + culture.

Agile	DevOps
• Focus on Dev	• End-to-End
• Iterative	• Continuous
• Team Centric	• Culture + Tools
4. Q. What is CI? **Ans:** Continuous Integration → Developers integrate code frequently, automated build & test run.
5. Q. What is CD? **Ans:** Continuous Delivery/Deployment → Code is always ready to release / auto deployed to production.
6. Q. What is the difference between CI&CD? **Ans:** CI = Integrate & Test
CD = Deliver & Deploy
7. Q. What is Infrastructure as Code (IaC)? **Ans:** Managing infrastructure using code (Terraform, CloudFormation, Pulumi, etc.).
8. Q. What is Automation in DevOps? **Ans:** Using tools/scripts to perform tasks automatically without manual intervention.
9. Q. What is a Pipeline? **Ans:** A series of automated steps to build, test, package and deploy the application.
10. Q. What is Version Control System? **Ans:** System that records changes to files over time (e.g., Git).
11. Q. What is a Repository? **Ans:** Storage location in VCS where code is stored (e.g., GitHub repo).
12. Q. What is Docker? **Ans:** Platform to containerize applications so they run consistently everywhere.
13. Q. What is Kubernetes? **Ans:** Container orchestration tool to deploy, scale and manage containers.
14. Q. What is configuration management? **Ans:** Managing system configurations using tools like Ansible, Puppet, Chef.
15. Q. What is Monitoring? **Ans:** Tracking system metrics, logs and alerts to ensure performance and availability.
16. Q. What is the role of Jenkins in DevOps? **Ans:** Jenkins is a CI/CD tool used to automate build, test and deployment.
17. Q. What is the purpose of Ansible? **Ans:** Configuration management and application deployment tool using simple YAML playbooks.
18. Q. What is meant by "Shift Left"? **Ans:** Testing and security moved early in the SDLC to find issues before production.
19. Q. What is a Container? **Ans:** Lightweight, standalone package with app + dependencies.
20. Q. What is Immutable Infrastructure? **Ans:** Infrastructure that is never modified after deployment; replace instead of change.

QUICK TIP

CI → Build & Test
CD → Release
CDE → Deploy

IMPORTANT POINTS

- ✓ Culture > Tools
- ✓ Automate Everything
- ✓ Measure Everything
- ✓ Collaborate Always
- ✓ Learn Continuously

REMEMBER

- People
- Process
- Tools
- Automation
- Measurement
- = Successful DevOps

COMMON INTERVIEW TOPICS

- ★ Linux Basics
- ★ Networking Basics
- ★ Cloud (AWS/Azure)
- ★ Git & GitHub
- ★ CI/CD Tools
- ★ Containers & K8s
- ★ IaC (Terraform)
- ★ Monitoring Tools
- ★ Scripting (Bash)

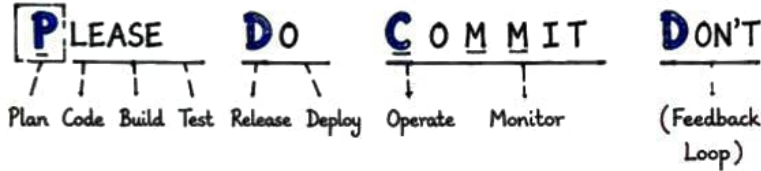


DEVOPS

CHEAT SHEET + MEMORY TRICKS

One Page
Revision

DEVOPS LIFECYCLE (MNEMONIC)



Or Simply Remember → **P C B T R D O M**

Plan → Code → Build → Test → Release → Deploy

→ Operate → Monitor (Continuous Feedback)

IMPORTANT KEYWORDS

- Automation
- Collaboration
- Continuous Integration (CI)
- Continuous Delivery (CD)
- Infrastructure as Code (IaC)
- Version Control
- Containerization
- Orchestration
- Monitoring
- Feedback Loop
- Scalability
- Reliability
- Security Shift-Left
- Idempotent
- Immutable Infrastructure

DEVOPS TOOLS QUICK MAP

	Git	→ Version Control System
	GitHub / GitLab	→ Code Hosting & Collaboration
	Jenkins	→ CI/CD Automation Server
	Docker	→ Containerization Platform
	Kubernetes	→ Container Orchestration
	Terraform	→ Infrastructure as Code
	Ansible	→ Configuration Automation
	Prometheus	→ Monitoring & Alerting
	Grafana	→ Visualization & Dashboards

QUICK INTERVIEW TIPS

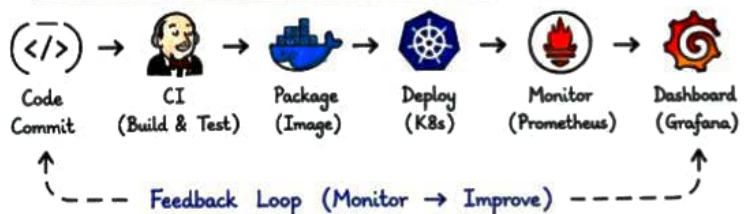
- ✓ Understand concepts before tools.
- ✓ Give real-world examples.
- ✓ Keep answers short, clear & structured.
- ✓ Highlight automation, monitoring and collaboration.
- ✓ Always end with how it adds value to business.
- ✓ Be honest, confident & calm 😊

DEVOPS PRINCIPLES (GOLDEN RULES) ☆

- ✓ Automate Everything Possible.
- ✓ Build Once, Deploy Anywhere.
- ✓ Monitor Everything, Improve Continuously.
- ✓ Fail Fast, Learn Fast.
- ✓ Collaboration over Communication.
- ✓ Security is Everyone's Responsibility.
- ✓ Keep It Simple, Scalable & Repeatable.



SHORT WORKFLOW (REVISION)



RAPID FIRE REVISION

- DevOps = Culture + Automation + Measurement
- CI = Integrate & Test Early
- CD = Deliver Fast & Deploy Often
- IaC = Manage Infra using Code
- Containers = Consistent Environment
- K8s = Orchestrate Containers at Scale
- Monitoring = Visibility
- Feedback Loop = Continuous Improvement
- Version Control = Track & Manage Changes
- Automation = Save Time, Reduce Errors
- Collaboration = High Performance Teams

MANTRA

★ Automate ★ Collaborate ★ Monitor ★ Improve ★ Deliver Value

Remember

Small changes done continuously make a **BIG** impact. 😊



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Keep Learning & Growing ❤️